

DTSA 5726

Introduction to Bayesian Statistics for Data Science

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Data Scientist (47.9 / 100)	38.9 / 100
SPECIALIZATION Bayesian Statistics for Data Science		

39

CURRENT AVG SCORE

across 6 job roles

72

POTENTIAL SCORE

with recommended additions

+44

MAX SCORE GAIN

best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces the theoretical, philosophical, and mathematical foundations of Bayesian Statistical inference. Students will learn to apply this foundational knowledge to real-world data science problems. Topics include the use and interpretations of probability theory in Bayesian inference; Bayes’ theorem for statistical parameters; conjugate, improper, and objective priors distributions; data science applications of Bayesian inference; and ethical implications of Bayesian statistics.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	47.9	64.2	+16.3	MEDIUM
Machine Learning Engineer	43.3	74.8	+31.5	HIGH
Applied Scientist	39.2	83.0	+43.8	HIGH
ML Engineer	39.2	72.2	+33.0	HIGH
AI Engineer	37.1	78.3	+41.2	HIGH
MLOps Engineer	26.7	56.6	+29.9	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Data Scientist	47.9 → 64.2 (+16.3 pts)
sets presentation	
Machine Learning Engineer	43.3 → 74.8 (+31.5 pts)
sets learning tools exploratory design development models machine	
Applied Scientist	39.2 → 83.0 (+43.8 pts)
computational research interdisciplinary models technical department learning industry	
ML Engineer	39.2 → 72.2 (+33.0 pts)
learning machine models new technical building tools	
AI Engineer	37.1 → 78.3 (+41.2 pts)
following models learning technical teams machine design new	
MLOps Engineer	26.7 → 56.6 (+29.9 pts)
learning machine tools models development error	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.